

1. 2012Q24

Which of the following descriptions about pollen grains is correct?

- A. Pollen grains are the male gametes of a plant.
- B. Pollen grains can produce nectar to attract insects.
- C. Pollen grains germinate when they land on the stigma of the same species.
- D. Pollen grains of insect-pollinated flowers are usually smaller than those of wind-pollinated flowers.

2. 2013Q14

Which of the following consists of individuals that are genetically different from each other?

- A. seeds of the same fruit
- B. stigmas of the same flower
- C. petals of the same plant
- D. buds of the same tuber

3. 2014Q3

Questions 3 and 4 refer to the following study:

A student wants to use an ordinary light microscope to observe the binary fission of a photosynthesizing protist under high magnification. A temporary mount of the protist is placed on the stage of the microscope.

Below are some steps in using a light microscope:

- (1) Focus with 10X objective
- (2) Focus with 40X objective
- (3) Search the field with 10X objective
- (4) Search the field with 40X objective
- (5) Move the slide until the protist is located in the centre of the field
- (6) Adjust light intensity if necessary

Which of the following is the most reasonable sequence of steps for the above study?

- A. (1), (3), (5), (6)
- B. (2), (6), (4), (5)
- C. (1), (2), (4), (5), (6)
- D. (1), (3), (5), (2), (6)

4. 2014Q4

Which of the following correctly describes binary fission of the protist?

- A. The amount of organelles in the daughter cell is the same as that in the mother cell.
- B. The number of chromosomes in the daughter cell is half of that in the mother cell.
- C. The alleles found in the daughter cells are different from each other.
- D. The sizes of the daughter cells are similar to each other.

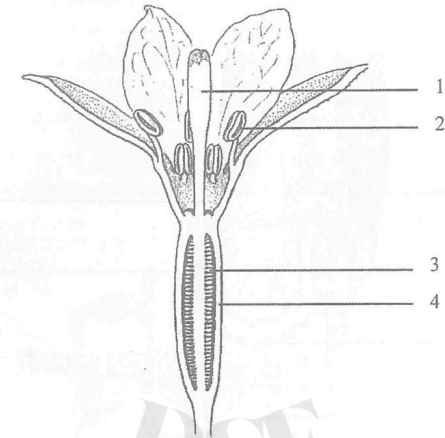
5. 2015Q19

Which of the following combinations correctly describes the absorption of water in the alimentary canal?

- (1) anchorage
- (2) absorption
- (3) vegetative propagation
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

6. 2015Q24

Question 24 refers to the diagram below, which shows the structures of a flower:

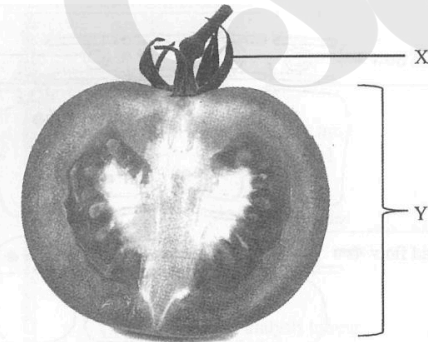


Which structure will develop into the fruit wall?

- A. 1
- B. 2
- C. 3
- D. 4

7. 2016Q31

Questions 31 and 32 refer to the diagram below, which shows the section of a fruit:



Which of the following combinations correctly shows the floral parts that develop into structure X and Y?

- | | X | Y |
|----|--------|-------|
| A. | carpel | petal |
| B. | carpel | ovary |
| C. | sepal | petal |
| D. | sepal | ovary |

8. 2016Q32

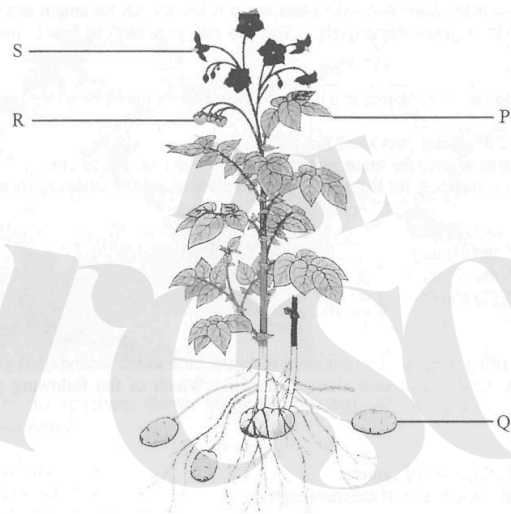
Structure Y serves as

- (1) food store for seed germination.
- (2) attraction to animals for seed dispersal.
- (3) cushion to protect the seed during falling.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

9. 2018Q15

Questions 15 to 17 refer to the following diagram, which shows the structures of a potato plant:



Which of the following labelled structure(s) is / are involved in the reproduction of this potato plant?

- A. Q only
- B. S only
- C. R and S only
- D. Q, R and S only

10. 2018Q16

Which of the following labelled structures contain cells with different genotypes?

- A. P and Q
- B. P and S
- C. Q and R
- D. R and S

11. 2018Q17

Farmers usually grow potato plants by vegetative propagation. This is probably because vegetative propagation

- A. produces tubers for harvesting.
- B. does not involve seed dispersal.
- C. takes a shorter time to reproduce.
- D. allows rapid colonization of an area.

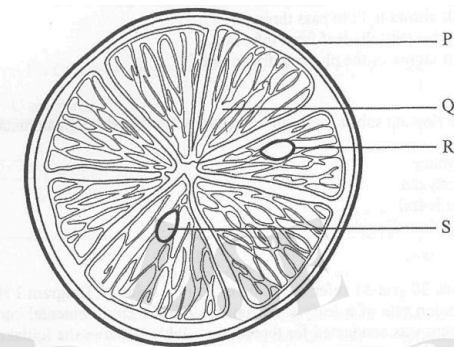
12. 2020Q25

Which of the following descriptions of floral structures is correct?

- A. The ovule forms seeds.
- B. The ovary forms seed coats
- C. Pollen grains are male gametes
- D. The filament contains pollen grains

13. 2022Q32

Questions 32 to 34 refer to the following diagram which shows a section of a fruit:



Which of the following structures is developed from the ovum?

- A. P
- B. Q
- C. R
- D. S

14. 2022Q33

Which of the following pairs of structures have the same genetic composition?

- (1) P and Q
 - (2) Q and R
 - (3) R and S
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

15. 2022Q34

Which of the following statements correctly describes the major role of Q?

- A. It protects the seeds by acting as a cushion.
- B. It provides nutrients to the seeds for germination.
- C. It helps seed dispersal by attracting animals to eat the fruit.
- D. It allows the seeds to survive through adverse conditions by storing food.

LQ

1. 2014Q8a, bi, bii

A primrose plant is a flowering plant that has two different types of flowers. Photograph I and Photograph II show the appearances of the two types of flowers (P and Q) and Photograph III and Photograph IV show the sections of the flowers respectively. Each primrose plant produces either type P flowers or type Q flowers.

Photograph I: Type P flower (x 4)



Photograph II: Type Q flower (x 4)



Photograph III: Section of type P flower (x 4)



Photograph IV: Section of type Q flower (x 4)

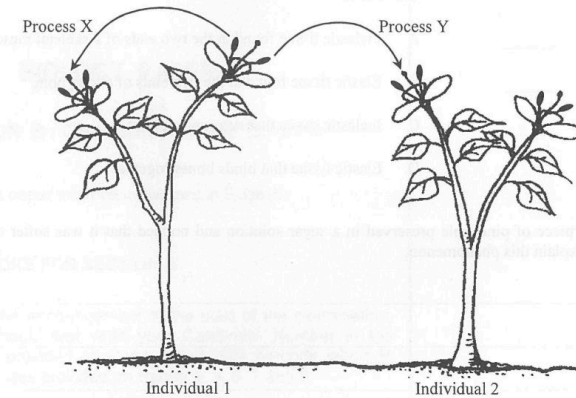


nectary

- a) Apart from the presence of nectaries, give **two** observable features of type P flowers which support the claim that the primrose is an insect-pollinated plant. (2 marks)
- b) Butterflies collect nectar from flowers using a mouth structure in the form of a long sucking tube.
 - i) When a butterfly visits a type P flower, which part of the sucking tube will the pollen grains stick to? (1 mark)
 - ii) When the same butterfly visits another flower, which type of flower will be more readily pollinated? Why? (2 marks)

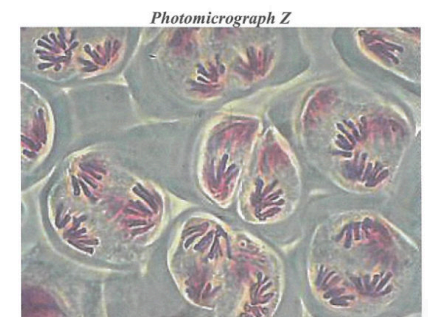
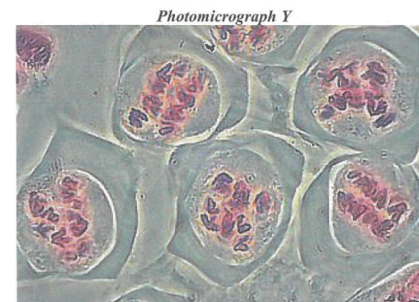
2. 2017Q3a, b

The diagram below shows certain processes occurring in a species of flowering plant:



- a) What is process Y? (1 mark)
- b) Describe the sequence of events leading to fertilisation after process Y has completed. (4 marks)

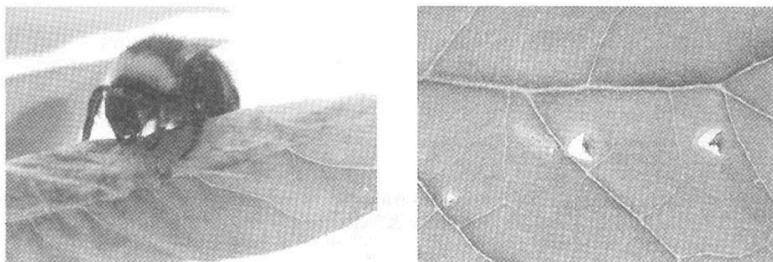
3. 2019Q3a



- a) State **one** floral structure in which this type of division takes place. (1 mark)

4. 2021Q7

- a) In flowering plants, environmental stress (i.e. under adverse conditions) in general can induce flowering. Explain why this flowering response can increase the chance of survival of flowering plants. (3 marks)
- b) Recently, scientists observed that bees cut tiny holes in leaves with their mouthparts (as shown in the photographs below) but did not consume or transport the leaf fragments:



It has been hypothesized that bees induce flowering by imposing a mechanical stress on the flowering plants. To test this hypothesis, three groups of tomato plants at the same developmental stage (without floral buds) were subjected to the following treatments respectively

1. Bees cut tiny holes in leaves (bee damage)
2. Similar holes in leaves were cut by using forceps (mechanical damage)
3. Intact leaves without treatment (no damage)

The time taken for flowering of each group of these tomato plants after the respective treatment was recorded.

- i) If the above hypothesis is correct, what would be the predicted results? (1 mark)
- ii) The table below shows the time taken for flowering of these tomato plants after the treatments:

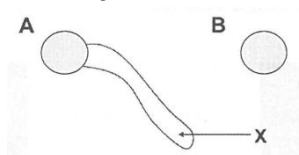
	Bee damage	Mechanical damage	No damage
Average time taken for flowering after the treatment (days)	38	56	70

With reference to the data shown in the table, discuss if the data support the hypothesis that bees induce flowering by imposing a mechanical stress on the flowering plants. (4 marks)

- c) When bees establish a new colony, they will inflict more leaf damage to the surrounding flowering plants if the colony is in an area with insufficient supply of pollen. What is the advantage of this behaviour to the bees? (1 mark)

5. 2024Q10b

In an investigation, pollen grains collected from a single flower were cultured in an artificial medium. After 48 hours incubation, they were observed under a light microscope. Two types of pollen grains with different appearances were observed, as shown in the diagram below:



- b) If these two types of pollen grains land on a stigma of the flower of the same species, which type of pollen grains will lead to formation of seed? Explain your answer. (3 marks)

Ans

2012Q24: C (37%)	2013Q14: A (80%)	2014Q3: D (59%)	2014Q4: D (41%)
2015Q19: D (40%)	2015Q24: D (78%)	2016Q31: D (70%)	2016Q32: B (30%)
2018Q15: D (61%)	2018Q16: D (51%)	2018Q17: C (59%)	2020Q25: A (57%)
2022Q32: D (64%)	2022Q33: C (33%)	2022Q34: C (63%)	

2014Q8a, bi, bii

- 8 (a) • coloured / large petals (1) and anther within the flower tube (1) (2)
- (b) (i) • middle part (1) (1)
- (ii) • type Q flower (1)
- because its stigma is located at the middle level of the flower tube (1) which is at the same position where the middle part of the sucking tube will touch upon (2)

2017Q3a, b

- 3 (a) • cross-pollination (1) (1)
- (b) • after landing on the stigma, pollens germinate to form pollen tubes (1)
- which carry the male gametes (1)
- pollen tubes grow along the style towards the ovary to reach the ovules / micropyle (1) (4)
- the male gametes will be released to fuse with the female gametes in the ovules (1)

2019Q3a

- 3 (a) • anther / ovary (1) (1)

2021Q7

- 7 (a) Concept for mark award:
- consequence of flowering (1)
 - explanation of how the chance of survival can be increased (any one of the following approaches)
 - dispersal of fruits / seeds (1) + description of how dispersal increases survival (1)
 - seeds can stay inactive (1) + description of how this increases survival (1)
 - dispersal of fruits / seeds (1) + seed can stay inactive
- (3)
- e.g.
- flowering results in the formation of fruits and seeds (1)
- (any **one** of the following sets)
- fruits / seeds are the dispersal units which allow the progeny / offspring to be dispersed away from the mother plants (1); thus the progeny / offspring can escape from the adverse conditions (1) and have a higher survival rate
 - the seeds can survive / remain inactive throughout the period with adverse conditions (1) and germinate when the conditions become favorable again (1)
 - fruits / seeds are the dispersal units which allow the progeny / offspring to be dispersed away (1) from the mother plants while the seeds can survive / remain inactive throughout the period of adverse conditions (1)
- (b) (i) • both bee damage and mechanical damage would shorten the flowering time to a similar extent (1) (1)
- (ii) Concept for mark award:
- comparison of effects of different treatments with the control (1) to draw valid conclusions (1) x2 (4)
- e.g.
- the time taken to flower in the plants with mechanical damage was shorter than in plants without damage (1), showing that mechanical damage can induce flowering (1)
 - however, bee damage advanced the flowering time a lot more than pure mechanical damage (1), which shows that there were factors other than mechanical damage which induced the flowering (1)
- (c) • It can ensure that there is a match between the timing of flower production and colony establishment / the bees can induce flowering when they are establishing their colonies / it can ensure that there is a sufficient food supply for the colony (1) (1)

9 marks

2024Q10b

	Concept for mark award	Example	Marks
10(b)	function of pollen tube	pollen tube / structure X transfers / carries / delivers the male gametes to the female gametes / egg / ovum inside the ovule	(1)
	fertilisation	the male gametes fertilise / fuse with the female gametes / ova in the ovule	(1)
	correct type of pollen tube	therefore, only type A pollen grains result in fertilisation that leads to the formation of seeds	(1)